Leonardo de Moura

Curriculum Vitae

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Homepage	`http://leodemoura.github.io`
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Scholar	`https://scholar.google.com/citations?user=CwazDKgAAAAJ&hl=en`
Nationality	American and Brazilian

Experience and Job History

04/2023 – present	Senior Principal Applied Scientist at AWS, Redmond, WA, USA
07/2023 – present	Chief Architect and co-founder Lean FRO (nonprofit), USA
01/2013 – 03/2023	Senior Principal Researcher at Microsoft, Redmond, WA, USA
01/2010 – 01/2013	Principal Researcher at Microsoft, Redmond, WA, USA
08/2006 – 01/2010	Senior Researcher at Microsoft, Redmond, WA, USA
02/2001 – 08/2006	Computer Scientist at SRI International, Menlo Park, CA, USA
04/2000 – 12/2000	Computer Engineer at Advus, São Paulo, Brazil
08/1994 – 04/2000	Research Assistant at the Software Engineering Laboratory of PUC-Rio
06/1998 – 12/1998	Visiting Researcher at Semantic Designs, Austin, TX, USA

Projects

I am the main architect of the following projects.

- *The Lean Proof Assistant and Programming Language*, `https://lean-lang.org`. Lean is implemented in Lean itself and is fully extensible: users can modify and extend the parser, elaborator, tactics, decision procedures, pretty printer, and code generator. Lean is also an efficient functional programming language based on a novel programming paradigm called functional but in-place. The project has been used by six Fields Medalists, and is featured in many popular science magazines, including **Nature** [1,2], **Wired** [3], **BigThink** [4], and **New York Times** [5]. Many additional links to articles and interviews are available at `https://lean-lang.org/fro/about`. The project has received numerous awards, including the *Programming Languages Software Award* from ACM SIGPLAN.
- *Z3* is an efficient satisfiability modulo theories (SMT) solver, `http://github.com/z3prover/z3`. Z3 is used in automated testing, software and hardware verification, optimization, and constraint solving applications. The project has received numerous awards, including the *Programming*

[1] `https://www.nature.com/articles/d41586-023-00487-2`
[2] `https://www.nature.com/articles/d41586-021-01627-2`
[3] `https://www.wired.com/story/the-effort-to-build-the-mathematical-library-of-the-future/`
[4] `https://bigthink.com/the-future/artificial-intelligence-replace-mathematicians/`
[5] `https://www.nytimes.com/2023/07/02/science/ai-mathematics-machine-learning.html`

Languages Software Award from ACM SIGPLAN. Z3 is widely used in industry (e.g., Microsoft, AWS, Apple, Meta, Google, to name a few).

Research Interests

Theorem Proving, Programming Languages, Model Checkers, Software/Hardware Verification, Decision Procedures, Static Analysis.

Awards

2025	Programming Languages Software Award for Lean from ACM SIGPLAN.
2025	Skolem Award for the paper "The Lean Theorem Prover". The Skolem award is given to the papers that have passed the test of time by being among the most influential in the field of automated deduction.
2021	Distinguished Paper Award at PLDI "Perceus: Garbage Free Reference Counting with Reuse".
2021	CAV Award for pioneering contributions to the foundations of the theory and practice of satisfiability modulo theories (SMT).
2019	Herbrand Award for numerous and important contributions to SMT solving, including its theory, implementation, and application to a wide range of academic and industrial needs.
2018	ETAPS 2018 Test of Time Award for the paper *Z3: An Efficient SMT Solver*.
2017	Skolem Award for the paper "Efficient E-Matching for SMT Solvers". The Skolem award is given to the papers that have passed the test of time by being among the most influential in the field of automated deduction.
2015	Programming Languages Software Award for Z3 from ACM SIGPLAN.
2014	TACAS Conference Award. Most influential tool paper in the first 20 years of TACAS.
2010	Haifa Verification Conference Award. The HVC award is given to the most influential work in the last five years in the scope of software and hardware verification and testing.
2007	Microsoft Gold Star (for the Z3 theorem prover) Microsoft
2005	SRI Focus Award (Outstanding Employee) SRI International
2000	Second Prize in the ACM'2000 Student Research Contest Association for Computing Machinery (ACM)

Education

04/2000 Ph.D. in Computer Science.
Thesis topic: “Automating the Generation of Program Analysis and Verification Tools.”
Pontifical Catholic University of Rio de Janeiro (PUC-Rio), Brazil.

03/1996 M.Sc. in Computer Science.
Thesis topic: “Visual Development Environments.”
Pontifical Catholic University of Rio de Janeiro (PUC-Rio), Brazil.

01/1994 Computer Engineer.
Pontifical Catholic University of Rio de Janeiro (PUC-Rio), Brazil.

Publications

1. K. Morrison and L. de Moura, *grind: An SMT-Inspired Tactic for Lean 4*, 13th International Joint Conference in Automated Reasoning (IJCAR), 2026.
2. C. Barrett, S. Chaudhuri, F. Montesi, J. Grundy, P. Kohli, L. de Moura, A. Rademaker and S. Yingchareonthawornchai, *CSLib: The Lean Computer Science Library*, arXiv:2602.04846, to appear in Communications of the ACM, 2026.
3. S. Ullrich and L. de Moura, *‘do’ unchained: Embracing local imperativity in a purely functional language*, Proc. ACM Program. Lang., ICFP, 2022.
4. L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, 28th International Conference on Automated Deduction, 2021.
5. A. Reinking, N. Xie, L. de Moura, and D. Leijen, *Perceus: Garbage free reference counting with reuse*, In Proceedings of the 42nd ACM SIGPLAN International Conference on Programming Language Design and Implementation, PLDI, 2021. (*Distinguished Paper Award*).
6. S. Ullrich and L. de Moura, *Beyond notations: Hygienic macro expansion for theorem proving languages*, 10th International Joint Conference in Automated Reasoning (IJCAR), 2020.
7. S. Ullrich and L. de Moura, *Counting immutable beans: Reference counting optimized for purely functional programming*, 31st Symposium on Implementation and Application of Functional Languages (IFP), 2019.
8. D. Selsam, M. Lamm, B. Bünz, P. Liang, L. de Moura and D. Dill, *Learning a SAT Solver from Single-Bit Supervision*, International Conference on Learning Representations (ICLR), 2019.
9. G. Ebner, S. Ullrich, J. Roesch, J. Avigad and L. de Moura, *A Metaprogramming Framework for Formal Verification*, Proc. ACM Program. Lang., ICFP, August 2017.
10. D. Selsam and L. de Moura, *Congruence Closure in Intensional Type Theory*, 8th International Joint Conference in Automated Reasoning (IJCAR), 2016.
11. R. Lewis and L. de Moura, *Automation and Computation in the Lean Theorem Prover*, International Conference on Artificial Intelligence and Theorem Proving (AITP), 2016.
12. J. Avigad, L. de Moura and S. Kong, *Theorem Proving in Lean*, 2015.
13. L. de Moura, S. Kong, J. Avigad, F. van Doorn and J. von Raumer, *The Lean Theorem Prover*, 25th International Conference on Automated Deduction, 2015.
14. C. Barrett, L. de Moura and P. Fontaine, *Proofs in Satisfiability Modulo Theories*, Mathematical Logic and Foundations. College Publications, London, UK, 2015.
15. A. Reynolds, C. Tinelli and L. de Moura, *Finding Conflicting Instances of Quantified Formulas in SMT*, 14th International Conference on Formal Methods in Computer-Aided Design, 2014.

16. D. Jovanović, C. Barrett, and L. de Moura, *The design and implementation of the model constructing satisfiability calculus*, 13th International Conference on Formal Methods in Computer-Aided Design, 2013.

17. L. de Moura and G. O. Passmore, *Computation in real closed infinitesimal and transcendental extensions of the rationals*, 24th International Conference on Automated Deduction, 2013.

18. L. de Moura and G. O. Passmore, *The Strategy Challenge in SMT Solving*, Automated Reasoning and Mathematics: Essays in Memory of William W. McCune, LNAI 7788, 2013.

19. L. de Moura, D. Jovanović, *A Model-Constructing Satisfiability Calculus*, 14th International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI), 2013.

20. D. Jovanović and L. de Moura, *Cutting to the chase solving linear integer arithmetic*, Journal of Automated Reasoning, 2013 (*submitted*).

21. G. Passmore, L. C. Paulson, L. de Moura, *Real algebraic strategies for MetiTarski proofs*, 11th International Conference, AISC 2012, 19th Symposium, Calculemus 2012.

22. D. Jovanović, L. de Moura, *Solving nonlinear arithmetic*, 6th International Joint Conference in Automated Reasoning (IJCAR), 2012.

23. D. Jovanović, L. de Moura, *Solving nonlinear arithmetic*, Technical Report MSR-TR-2012-20, Microsoft Research, 2012.

24. C. Barrett, M. Deters, L. de Moura, A. Oliveras, and A. Stump, *6 Years of SMT-COMP*, Journal of Automated Reasoning, 2012.

25. N. Bjorner and L. de Moura, *Tractability and Modern Satisfiability Modulo Theories Solvers*, Handbook of Tractability, Cambridge University Press, 2012.

26. K. Hoder, N. Bjorner, and L. de Moura, *μZ – an efficient engine for fixed points with constraints*, Computer Aided Verification (CAV), 2011.

27. L. de Moura and N. Bjorner, *Satisfiability modulo theories: introduction and applications*, Communications of the ACM (CACM), 2011.

28. D. Jovanović and L. de Moura, *Cutting to the chase solving linear integer arithmetic*, 23rd International Conference on Automated Deduction (CADE), 2011.

29. M. P. Bonacina, C. Lynch, and L. de Moura, *On deciding satisfiability by theorem proving with speculative inferences*, Journal of Automated Reasoning, 2011.

30. M. Veanes, N. Bjorner and L. de Moura, *Symbolic Automata Constraint Solving*, International Conference on Logic programming and automated reasoning (LPAR), 2010.

31. C. Wintersteiger, Y. Hamadi and L. de Moura, *Efficiently Solving Quantified Bit-Vector Formulas*, International Conference on Formal Methods in Computer-Aided Design (FMCAD), 2010.

32. L. de Moura and N. Bjorner, *Bugs, Moles and Skeletons: Symbolic Reasoning for Software Development*, International Joint Conference on Automated Reasoning (IJCAR), 2010.

33. N. Bjorner and L. de Moura, *TAPAS Theory Combinations and Practical Applications*, invited paper at FORMATS, 2009.

34. L. de Moura and N. Bjorner, *Generalized and Efficient Array Decision Procedures*, International Conference on Formal Methods in Computer-Aided Design (FMCAD), 2009.

35. L. de Moura and N. Bjorner, *Satisfiability Modulo Theories: An Appetizer*, invited paper to SBMF, 2009.

36. G. O. Passmore and L. de Moura, *Superfluous S-polynomials in Strategy-Independent Grobner Bases*, 11th International Symposium on Symbolic and Numeric Algorithms for Scientific Computing (SYNASC), 2009.

37. L. de Moura and G. O. Passmore, *On Locally Minimal Nullstellensatz Proofs*, International Workshop on Satisfiability Modulo Theories (SMT), 2009.

38. G. O. Passmore and L. de Moura, *Universality of Polynomial Positivity and a Variant of Hilbert's 17th Problem*, ADDCT'09.

39. L. de Moura and N. Bjorner, *Z*3^{10}*: Applications, Enablers, Challenges and Directions*, invited paper to CFV, 2009.

40. M. P. Bonacina, C. Lynch and L. de Moura, *On deciding satisfiability by DPLL(*$\Gamma + T$*) and unsound theorem proving*, 22nd International Conference on Automated Deduction (CADE-22), 2009.

41. Y. Ge and L. de Moura, *Complete instantiation for quantified SMT formulas*, International Conference on Computer Aided Verification (CAV), 2009.

42. C. Wintersteiger, Y. Hamadi and L. de Moura, *A Concurrent Portfolio Approach to SMT Solving*, International Conference on Computer Aided Verification (CAV), 2009.

43. R. Piskac, L. de Moura and N. Bjorner, *Deciding Effectively Propositional Logic with Equality*, Technical Report MSR-TR-2008-181.

44. N. Bjorner, B. Dutertre and L. de Moura, *Accelerating Lemma Learning using Joins – DPLL(Join)*, International Conference on Logic programming and automated reasoning (LPAR), 2008.

45. L. de Moura and N. Bjorner, *Proofs and Refutations, and Z3*, IWIL, 2008.

46. N. Bjorner, L. de Moura and N. Tillmann, *Satisfiability Modulo Bit-precise Theories for Program Exploration*, Invited workshop paper, CFV, 2008.

47. L. de Moura and N. Bjorner, *Deciding Effectively Propositional Logic using DPLL and substitution sets*, International Joint Conference on Automated Reasoning (IJCAR), 2008.

48. L. de Moura, N. Bjorner, *Engineering DPLL(T) + Saturation*, International Joint Conference on Automated Reasoning (IJCAR), Sydney, Australia, 2008.

49. L. de Moura and N. Bjorner, *Z3: An Efficient SMT Solver*, International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS), 2008.

50. L. de Moura and N. Bjorner, *Relevancy Propagation*, MSR Technical Note, 2007.

51. L. de Moura and N. Bjorner, *Efficient E-matching for SMT solvers*, International Conference on Automated Deduction (CADE), 2007.

52. L. de Moura and N. Bjorner, *Model-based Theory Combination*, Workshop on Satisfiability Modulo Theories (SMT), 2007.

53. C. Barrett, L. de Moura and A. Stump, *Design and Results of the Second Satisfiability Modulo Theories Competition (SMT-COMP 2006)*, Journal of Formal Methods in System Design, 2007.

54. L. de Moura, B. Dutertre and N. Shankar, *A Tutorial on Satisfiability Modulo Theories*, Conference on Computer Aided Verification (CAV), 2007.

55. B. Dutertre and L. de Moura, *A Fast Linear-Arithmetic Solver for DPLL(T)*, 18th International Conference on Computer Aided Verification (CAV'06).

56. C. Barrett, L. de Moura and A. Stump, *Design and Results of the 1st Satisfiability Modulo Theories Competition (SMT-COMP 2005)*, Journal of Automated Reasoning (JAR), 2006.

57. C. Barrett, L. de Moura and A. Stump, *SMT-COMP: Satisfiability Modulo Theories Competition*, 17th International Conference on Computer Aided Verification (CAV'05).
58. G. Hamon, L. de Moura and J. Rushby, *Generating Efficient Test Sets with a Model Checker*, The Second IEEE International Conference on Software Engineering and Formal Methods (SEFM'04).
59. L. de Moura, H. Rueß and N. Shankar, *Justifying Equality*, Second Workshop on Pragmatics of Decision Procedures in Automated Reasoning (PDPAR'04).
60. L. de Moura, S. Owre, H. Rueß, J. Rushby, N. Shankar, M. Sorea and A. Tiwari, *SAL 2*, 16th International Conference on Computer Aided Verification (CAV'04).
61. L. de Moura and H. Rueß, *An Experimental Evaluation of Ground Decision Procedures*, 16th International Conference on Computer Aided Verification (CAV'04).
62. L. de Moura, H. Rueß, N. Shankar and J. Rushby, *The ICS decision procedures for embedded deduction*, Second International Joint Conference on Automated Reasoning (IJCAR'04).
63. H. Rueß and L. de Moura, *From Simulation to Verification (and Back)*, Proceedings of the 2003 Winter Simulation Conference.
64. L. de Moura, H. Rueß, J. Rushby and N. Shankar, *Embedded Deduction with ICS*, Presented at the third High Confidence Software and Systems Conference, 2003.
65. L. de Moura, H. Rueß and M. Sorea, *Bounded Model Checking and Induction: From Refutation to Verification*, 15th International Conference on Computer Aided Verification (CAV'03).
66. L. de Moura, H. Rueß and M. Sorea, *Lazy Theorem Proving for Bounded Model Checking over Infinite Domains*, International Conference on Automated Deduction (CADE'02).
67. L. de Moura, C. J. P. de Lucena and E. H. Haeusler, *Analysis of Parallel Programs*, Electronic Notes in Theoretical Computer Science, 2002.
68. L. de Moura and H. Rueß, *Lemmas on Demand for Satisfiability Solvers*, Fifth International Symposium on the Theory and Applications of Satisfiability Testing (SAT), 2002.
69. L. de Moura, *Semantic-Directed Generation of Program Analysis and Verification Tools*, Second Prize in the ACM 2000 Student Research Contest, Austin, Texas, 2000.
70. L. de Moura, C. J. P. de Lucena and E. H. Hausler, *Analysis of Parallel Programs*, Brazilian Symposium of Programming Languages (SBLP), 2000.
71. L. de Moura, C. J. P. de Lucena and E. H. Hausler, *A Modular Implementation of Action Notation*, International Workshop on Action Semantics and Related Frameworks, 2000.
72. M. F. Fontoura, C. Braga, L. de Moura and C. J. P. de Lucena, *Using Domain Specific Languages to Instantiate Object-Oriented Frameworks*, IEE Proceedings – Software, 147(4), 2000.
73. M. F. Fontoura, L. de Moura, S. Crespo and C. J. P. de Lucena, *ALADIN: An Architecture for Learningware Application Design and Instantiation*, World Wide Web WWW Baltzer Science, Bussum, Holland, 2000.
74. I. D. Baxter, A. Yahin, S. Nedunuri, and L. de Moura, *Lowering Maintenance Costs by Code Clone Removal*, 12th International Software Quality Week, 1999.
75. L. de Moura, C. J. P. de Lucena and A. von Staa, *The Spider Environment*, Software Practice & Experience, 29(2), 99–124, 1999.
76. I. D. Baxter, A. Yahin, L. de Moura, M. Sant'Anna and L. Bier, *Clone Detection Using Abstract Syntax Trees*, Proc. of the International Conference on Software Maintenance '98, 1998, IEEE Press.

77. L. de Moura and C. J. P. de Lucena, *An Introduction To The Spider Visual Programming Environment*, Brazilian Symposium of Software Engineering (SBES), 1997.

78. H. Fuks and L. de Moura, *Supporting Team Collaboration*, SIGOIS Bulletin 16, New York, pp. 64–68, 1995.

79. H. Fuks and L. de Moura, *A Document Based Approach for Cooperation*, Journal of the Brazilian Computer Society, V1, N1, pp. 36–45, July 1994.

80. L. de Moura and R. R. dos Santos, *Critical Exponents for Site-Bond Correlated Percolation*, Phys. Review B 45, 1023, 1992.

Patents

- L. de Moura and N. Bjorner, *Matching based pattern inference for SMT solvers*, US Patent 9,489,221.
- L. de Moura and N. Bjorner, *Relevancy propagation for efficient theory combination*, US Patent 8,140,459.
- L. de Moura and N. Bjorner, *E-matching for SMT solvers*, US Patent 8,103,674.
- L. de Moura and N. Bjorner, *Model-based theory combination*, US Patent 7,925,476.
- J. Rushby, L. de Moura, G. Hamon, *Formal methods for test case generation*, US Patent 7,865,339.
- L. de Moura and H. Rueß, *Method for combining decision procedures with satisfiability solvers*, US Patent 7,653,520.

References

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